

Theorem 2 (RS-achievable message scaling). *Let $S = S(m)$ be the maximum Hamming weight and fix $E_2 \in (0, 1/2)$. For even n , choose*

$$r = \frac{n}{2}, \quad L = 2^{n/2}. \quad (1)$$

Then the RS tagging construction supports any admissible message length $m = rK$ satisfying

$$mS(m) \leq \frac{n}{2} 2^{n(\frac{1}{2} - E_2)}. \quad (2)$$

Specifically, it yields an

$$\left(n, m, \mathcal{F}_m^W(\leq S(m)), 0, 2^{-nE_2} \right) \quad (3)$$

BFC code. Consequently, for a given scaling function $\Phi(R, n)$, an RS-achievable rate is obtained by substituting $m = \Phi(R, n)$ into (2).

Proof. By Theorem 1,

$$\lambda_2 \leq \frac{S(m)(K-1)}{L} \leq \frac{S(m)K}{L} = \frac{mS(m)}{rL}. \quad (4)$$

Substituting $r = n/2$, $L = 2^{n/2}$, and (2) gives $\lambda_2 \leq 2^{-nE_2}$. □